Bermudan Swaptions in One- and Two-Factor HJM Models

Calibration, PDE, Grid and Monte Carlo Pricing, and a Review of the Model

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Abstract

We price co-terminal Bermudan swaptions in the one-factor Heath–Jarrow–Morton model with separable volatility $\sigma_f(t,T) = \sigma(t)e^{-\lambda(T-t)}$. The model is calibrated exactly to co-terminal European swaptions quoted in normal volatility, by a sequential bootstrap built on Jamshidian's decomposition, and the Bermudan is priced by a finite-difference scheme on the Markov factor and by Longstaff–Schwartz Monte Carlo. Reviewing the original implementation, we show that its zero-coupon bond formula carries the convexity adjustment with the wrong sign: the model then violates the martingale property of discounted bonds by 1.8 % on a 7-year bond, prices the same European swaption at 0.0258 by Jamshidian and 0.0206 by the PDE, and undervalues the reference Bermudan by 11 %. We derive the correct formula, measure the other numerical issues (Monte Carlo discounting, regression bias, time scheme), and study how the mean-reversion parameter, which the calibration cannot identify, drives the Bermudan price through the correlation between exercise dates. Finally we add a second factor (section 10): Jamshidian's decomposition is lost and replaced by two-dimensional integrals and a two-dimensional backward induction. On the complete EUR market of 30 March 2016 (240 at-the-money swaptions, 16 caps, smiles) the second factor divides the root-mean-square error on the fitted swaptions by 3.2 (12.4 to 3.9 bp) and on the fitted caps by 3.6, moves the Bermudan by −3.8 %, and, with the Europeans held fixed, the correlation alone moves the Bermudan by 12 %. All computations are reproduced by the interactive page https://yassinebarnicha.github.io/swaptions-hjm/.

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1 Introduction

A Bermudan swaption gives its holder the right to enter, on any of a set of dates $T_1 < \cdots < T_m$, an interest-rate swap that ends at a fixed final date T_N. Each exercise right alone is a European swaption, which the market quotes; the Bermudan adds the right to *choose* the date, and the value of that choice depends on how rates at the different dates are correlated, which European prices do not reveal. A pricing model must therefore (i) reproduce the Europeans, (ii) have a realistic dynamics across dates, and (iii) be numerically tractable for an optimal stopping problem.

The one-factor Gaussian HJM model with separable volatility (equivalently the Hull–White model with time-dependent volatility) meets the third requirement best: it is Markov in one Gaussian factor, bonds are exponential-affine in it, Europeans have closed forms, and the Bermudan reduces to a one-dimensional PDE. Its mean-reversion speed λ sets the inter-date correlation.

Contributions. This report documents the project and its review:

1. a self-contained derivation of the model (section 2), of the European pricing formulas (section 3) and of the calibration (section 4);
2. the PDE and Monte Carlo methods for the Bermudan, with their error analysis and convergence study (sections 5 to 7);

3. the model review (section 8): a sign error in the bond formula, exposed by a martingale test, and its impact;
4. a study of mean reversion under recalibration (section 9).

2 The model

2.1 The HJM framework

Let $P(t,T)$ be the price at t of the zero-coupon bond paying 1 at T, $f(t,T) = -\partial_T \ln P(t,T)$ the instantaneous forward rate, $r_t = f(t,t)$ the short rate and $\beta_t = \exp\int_0^t r_s\,\mathrm{d}s$ the bank account. Under the risk-neutral measure $\mathbb{Q}$, with a one-dimensional Brownian motion W, HJM specify

$$\mathrm{d}f(t,T) = \mu(t,T)\,\mathrm{d}t + \sigma_f(t,T)\,\mathrm{d}W_t, \qquad f(0,T) = f^{\mathrm{mkt}}(0,T). \tag{1}$$

Proposition 1 (HJM drift condition)**.** *Discounted bond prices $P(t,T)/\beta_t$ are $\mathbb{Q}$-martingales for all T if and only if*

$$\mu(t,T) = \sigma_f(t,T)\int_t^T \sigma_f(t,u)\,\mathrm{d}u. \tag{2}$$

Proof. Let $\Sigma(t,T) = \int_t^T \sigma_f(t,u)\,\mathrm{d}u$ and $A(t,T) = \int_t^T \mu(t,u)\,\mathrm{d}u$. From $\ln P(t,T) = -\int_t^T f(t,u)\,\mathrm{d}u$ and the stochastic Fubini theorem, $\mathrm{d}\ln P(t,T) = \big(r_t - A(t,T)\big)\,\mathrm{d}t - \Sigma(t,T)\,\mathrm{d}W_t$, so $\mathrm{d}\big(P/\beta\big)/(P/\beta) = \big(\tfrac12\Sigma^2 - A\big)\,\mathrm{d}t - \Sigma\,\mathrm{d}W_t$. The drift vanishes for all T iff $A(t,T) = \tfrac12\Sigma(t,T)^2$; differentiating in T gives (2). □

2.2 Separable volatility and the Markov factor

The model uses

$$\sigma_f(t,T) = \sigma(t)\,e^{-\lambda(T-t)} = \eta(t)\,b(T), \qquad \eta(t) = \sigma(t)e^{\lambda t}, \quad b(T) = e^{-\lambda T}, \tag{3}$$

with $\sigma(\cdot)$ piecewise constant on the exercise grid: $\sigma(t) = \sigma_k$ for $t \in (T_{k-1}, T_k]$, $T_0 = 0$. Define

$$\begin{aligned} X_t &= \int_0^t \eta(u)\,\mathrm{d}W_u, \qquad & \alpha(t) &= \int_0^t \eta(u)^2\,\mathrm{d}u, \\ \bar B(t) &= \int_0^t b(u)\,\mathrm{d}u = \frac{1-e^{-\lambda t}}{\lambda}, \qquad & \gamma(t) &= \int_0^t \eta(u)^2\,\bar B(u)\,\mathrm{d}u. \end{aligned} \tag{4}$$

X is a continuous Gaussian martingale, $X_t \sim \mathcal{N}(0,\alpha(t))$, with independent increments. On a segment $[a,b]$ where $\sigma \equiv \sigma_j$,

$$\alpha(b) - \alpha(a) = \sigma_j^2\,\frac{e^{2\lambda b} - e^{2\lambda a}}{2\lambda}, \qquad \gamma(b) - \gamma(a) = \frac{\sigma_j^2}{\lambda}\left[\frac{e^{2\lambda b} - e^{2\lambda a}}{2\lambda} - \frac{e^{\lambda b} - e^{\lambda a}}{\lambda}\right], \tag{5}$$

with the limits $\sigma_j^2(b-a)$ and $\tfrac12\sigma_j^2(b^2-a^2)$ when $\lambda \to 0$.

Proposition 2 (Forward rates, short rate and bonds)**.** *For $t \le T$,*

$$f(t,T) = f(0,T) + b(T)\big[\bar B(T)\,\alpha(t) - \gamma(t) + X_t\big], \tag{6}$$

$$r_t = f(0,t) + e^{-\lambda t}\big[\bar B(t)\,\alpha(t) - \gamma(t)\big] + e^{-\lambda t}X_t, \tag{7}$$

$$P(t,T) = \frac{P(0,T)}{P(0,t)}\exp\Big(-D(t,T)\big[c(t,T) + X_t\big]\Big), \tag{8}$$

where $D(t,T) = \bar B(T) - \bar B(t)$ and $c(t,T) = \tfrac12\big(\bar B(T) + \bar B(t)\big)\alpha(t) - \gamma(t)$.

Proof. By (2), $\mu(s,T) = \eta(s)^2 b(T)\big(\bar B(T) - \bar B(s)\big)$. Integrating $\mathrm{d}f(s,T)$ over $s \in [0,t]$,

$$f(t,T) = f(0,T) + b(T)\int_0^t \eta(s)^2\big(\bar B(T) - \bar B(s)\big)\,\mathrm{d}s + b(T)\int_0^t \eta(s)\,\mathrm{d}W_s,$$

which is (6); (7) is $T = t$. For the bond, $\int_t^T b(u)\,\mathrm{d}u = D(t,T)$ and $\int_t^T b(u)\bar B(u)\,\mathrm{d}u = \frac12(\bar B(T)^2 - \bar B(t)^2) = \frac12 D(t,T)(\bar B(T) + \bar B(t))$, so $\int_t^T \big[f(t,u) - f(0,u)\big]\,\mathrm{d}u = D(t,T)\big[c(t,T) + X_t\big]$ and (8) follows from $P(0,T)/P(0,t) = \exp\big(-\int_t^T f(0,u)\,\mathrm{d}u\big)$. □

Remark 1 (Hull–White). Applying Itô's formula to (7) gives $\mathrm{d}r_t = \big(\theta(t) - \lambda r_t\big)\,\mathrm{d}t + \sigma(t)\,\mathrm{d}W_t$ for a deterministic $\theta(\cdot)$ that fits the initial curve: the model is the Hull–White model with time-dependent volatility. λ is the speed of mean reversion of the short rate and $\sigma(t)$ its normal volatility.

Remark 2 (Dynamics of bond ratios). From (8), the diffusion part of $\ln P(t,T)$ is $-D(t,T)\,\eta(t)\,\mathrm{d}W_t$, hence that of $\ln\big(P(t,T_i)/P(t,T_e)\big)$ is $-\big(\bar B(T_i) - \bar B(T_e)\big)\eta(t)\,\mathrm{d}W_t$: a deterministic constant times $\mathrm{d}X_t$. Its integrated variance on $[0,T_e]$ is $D(T_e,T_i)^2\,\alpha(T_e)$.

3 European swaptions

3.1 Swaps and market quotes

A payer swap starting at T_0 with fixed rate K, payment dates $T_1 < \cdots < T_n$ and accruals τ_i is worth, at T_0, $1 - P(T_0,T_n) - K\sum_i \tau_i P(T_0,T_i)$ per unit notional. With the annuity $A_t = \sum_i \tau_i P(t,T_i)$ and the forward swap rate $S_t = \big(P(t,T_0) - P(t,T_n)\big)/A_t$, the swap value is $A_t(S_t - K)$ and the payer swaption pays $A_{T_0}(S_{T_0} - K)^+$. Under the annuity measure $\mathbb{Q}^A$ (numéraire A), S is a martingale. The market convention (Bachelier) takes $\mathrm{d}S_t = \sigma_N\,\mathrm{d}W_t^A$, so that

$$V_0^{\mathrm{pay}} = A_0\,\mathbb{E}^A\big[(S_{T_0} - K)^+\big] = A_0\Big[(S_0 - K)\,\Phi(d) + \sigma_N\sqrt{T_0}\,\varphi(d)\Big], \qquad d = \frac{S_0 - K}{\sigma_N\sqrt{T_0}}, \tag{9}$$

and $V_0^{\mathrm{rec}} = A_0\big[(K - S_0)\Phi(-d) + \sigma_N\sqrt{T_0}\,\varphi(d)\big]$. A quote σ_N is thus a price.

3.2 Options on zero-coupon bonds

Proposition 3. *The price at* 0 *of a put with expiry* T_e *and strike* K *on the bond* $P(\cdot,T_i)$, $T_i > T_e$, *is*

$$\begin{aligned} \mathrm{ZBP}(T_e,T_i,K) &= K\,P(0,T_e)\,\Phi(-h+v) - P(0,T_i)\,\Phi(-h), \\ v &= D(T_e,T_i)\sqrt{\alpha(T_e)}, \qquad h = \frac{1}{v}\ln\frac{P(0,T_i)}{K\,P(0,T_e)} + \frac{v}{2}, \end{aligned} \tag{10}$$

and the call is $\mathrm{ZBC} = P(0,T_i)\Phi(h) - K\,P(0,T_e)\Phi(h-v)$.

Proof. Under the T_e-forward measure, $Y_t = P(t,T_i)/P(t,T_e)$ is a martingale; by remark 2 it is lognormal with total variance v^2 at T_e, and $Y_{T_e} = P(T_e,T_i)$. Then $\mathrm{ZBP} = P(0,T_e)\,\mathbb{E}^{T_e}[(K - Y_{T_e})^+]$ is Black's formula with forward $Y_0 = P(0,T_i)/P(0,T_e)$. □

3.3 Jamshidian's decomposition

Proposition 4 (Jamshidian)**.** *Let* $c_i = K\tau_i$ *for* $i < n$ *and* $c_n = 1 + K\tau_n$. *There is a unique* $x^\star$ *with* $\sum_i c_i P(T_0,T_i;x^\star) = 1$, *and with* $K_i = P(T_0,T_i;x^\star)$,

$$V_0^{\mathrm{pay}} = \sum_{i=1}^n c_i\,\mathrm{ZBP}(T_0,T_i,K_i), \qquad V_0^{\mathrm{rec}} = \sum_{i=1}^n c_i\,\mathrm{ZBC}(T_0,T_i,K_i). \tag{11}$$

Proof. The payer payoff is $\left(1-\sum_i c_i P(T_0,T_i;X_{T_0})\right)^+$. By (8), each $P(T_0,T_i;x)$ is strictly decreasing in x ($D>0$), from $+\infty$ to 0, so $\sum_i c_i P(T_0,T_i;x)$ crosses 1 exactly once, at $x^\star$. For $x \geq x^\star$ every $P(T_0,T_i;x) \leq K_i$ and for $x < x^\star$ every $P(T_0,T_i;x) > K_i$; in both cases $\left(1-\sum_i c_i P_i\right)^+ = \sum_i c_i (K_i - P_i)^+$, using $\sum_i c_i K_i = 1$. Take expectations and apply proposition 3. □

The implementation finds $x^\star$ with Brent's method on $[-10, 10]$. Put–call parity (payer − receiver = forward swap) holds to 10^{-8}.

4 Calibration

4.1 The co-terminal bootstrap

Each exercise date T_k carries the co-terminal European swaption $T_k \to T_N$, quoted at $\sigma_{N,k}$. By propositions 3 and 4, its model price depends on $\sigma(\cdot)$ only through $\alpha(T_k)$, which by (5) depends on $\sigma_1, \ldots, \sigma_k$ only. Each bond put increases with v, so the price is continuous and strictly increasing in σ_k once $\sigma_1, \ldots, \sigma_{k-1}$ are fixed.

Proposition 5. *Given λ and $\sigma_1, \ldots, \sigma_{k-1}$, the equation $V_k^{\text{model}}(\sigma_k) = V_k^{\text{Bach}}(\sigma_{N,k})$ has at most one solution $\sigma_k > 0$; it exists iff the market price exceeds the model price with $\sigma_k = 0$, i.e. iff the market does not imply less total variance at T_k than is already accumulated on $[0, T_{k-1}]$.*

The bootstrap solves the dates in increasing order with Brent's method on $[\sigma_{\min}, \sigma_{\max}]$, and the fit is exact (residuals below 10^{-10}). When the condition of proposition 5 fails, the page reports the instrument that cannot be fitted instead of returning a wrong σ.

4.2 Mean reversion is not identified

For every λ in a wide range, proposition 5 produces a $\sigma(\cdot)$ that reproduces all the co-terminal Europeans exactly. The calibration set therefore leaves a one-parameter family of perfectly calibrated models, whose Bermudan prices differ (section 9): on our reference case, from 0.01656 ($\lambda = 0$) to 0.01848 ($\lambda = 0.3$), a spread of 12 %. λ must be set from other information: historical correlation of rates, or other instruments (caps, diagonal or column swaptions) whose prices depend on inter-date correlation.

5 The Bermudan swaption

The exercise value at T_k is the underlying co-terminal swap, $h_k(x) = \left(1 - P(T_k,T_N;x) - K\sum_{T_i>T_k} \tau_i P(T_k,T_i;x)\right)^+$ for a payer. The price is the value of an optimal stopping problem over the stopping times τ taking values in $\{T_1, \ldots, T_m\}$:

$$V_0 = \sup_\tau \mathbb{E}^{\mathbb{Q}}\left[\beta_\tau^{-1} h_\tau(X_\tau)\right]. \tag{12}$$

Since X is Markov, dynamic programming gives

$$V_{T_m} = h_m, \qquad V_{T_k} = \max\left(h_k, C_k\right), \qquad C_k(x) = \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_{T_k}^{T_{k+1}} r_s\,\mathrm{d}s} V_{T_{k+1}} \,\Big|\, X_{T_k} = x\Big]. \tag{13}$$

Exercising always at a fixed T_k is admissible, so $V_0 \geq \max_k V_k^{\text{Eur}}$; the difference is the *early-exercise* (or switch) premium. On the reference case below the Bermudan is worth 0.016746 against 0.014328 for the most expensive European.

Reference case. Curve $P(0,t)$ at $t = 1, 2, 3, 5, 7, 10$: 0.978, 0.954, 0.930, 0.885, 0.842, 0.780; payer, $K = 3\,\%$, annual payments from 2 to $T_N = 7$; exercise dates $1, 2, 3$ with normal vols 80, 85, 90 bp; $\lambda = 3\,\%$. The bootstrap gives $\sigma = (86.12,\ 96.45,\ 106.37)$ bp.

6 The PDE method

6.1 Equation and boundary conditions

Between exercise dates, by Feynman–Kac, $V(t,x) = \mathbb{E}[e^{-\int_t^T r} V(T, X_T) \mid X_t = x]$ solves

$$\partial_t V + \tfrac{1}{2}\eta(t)^2\, \partial_{xx} V - r(t,x)\, V = 0, \qquad r(t,x) = A(t) + e^{-\lambda t} x, \tag{14}$$

with $A(t) = f(0,t) + e^{-\lambda t}[\bar{B}(t)\alpha(t) - \gamma(t)]$ from (7). There is no drift term because X is a martingale; this is why the factor X, rather than r, is the natural state variable. At each T_k the value is updated by $V \leftarrow \max(V, h_k)$, and the equation is solved backwards from $V(T_m, \cdot) = h_m$.

The domain is truncated to $[-n\sqrt{\alpha(T_m)},\, n\sqrt{\alpha(T_m)}]$. Far from the origin the option is either deep in the money (exercise immediately) or worthless, so Dirichlet values equal to the largest intrinsic value among the remaining exercise dates are imposed. The truncation error is of the order of the probability of leaving the domain, $2\Phi(-n) \approx e^{-n^2/2}$, which is negligible from $n = 5$ (table 1).

6.2 Discretisation

On a uniform grid x_j with step Δx, let $\mathcal{L}^n$ be the discrete operator $(\mathcal{L}^n V)_j = \tfrac{1}{2}\eta(t_n)^2\,(V_{j+1} - 2V_j + V_{j-1})/\Delta x^2 - r(t_n, x_j)V_j$. The θ-scheme reads

$$\frac{V^n - V^{n+1}}{\Delta t} = \theta\, \mathcal{L}^n V^n + (1-\theta)\, \mathcal{L}^{n+1} V^{n+1}, \tag{15}$$

a tridiagonal system per step, solved by the Thomas algorithm in $O(n_x)$ operations. The time grid contains every exercise date, so that the max is applied exactly.

- $\theta = 1$ (implicit Euler, the original code) is unconditionally stable and monotone whenever $1 + \Delta t\, r(t,x) > 0$, but only first order in time.
- $\theta = \tfrac{1}{2}$ (Crank–Nicolson) is second order and unconditionally stable in L^2, but not L-stable: the high-frequency components created by the kink of h_k at each exercise date are damped by a factor close to -1 per step and oscillate.
- *Rannacher time-stepping* replaces the first two Crank–Nicolson steps after each exercise date by implicit steps (here: two full steps). This damps the oscillations and restores second-order convergence for non-smooth data (Rannacher 1984; Giles and Carter 2006).

6.3 Convergence

Figure 1 shows the error in time against a reference computed with $n_t = 12\,800$, and the error in space. Implicit Euler converges at order one and Crank–Nicolson with Rannacher steps at order two (the error is divided by about 4 to 5 each time n_t doubles). On this price, plain Crank–Nicolson is even more accurate at coarse steps: the oscillations it leaves near the exercise boundary are small and partly cancel in the value at $x = 0$; they matter more for the Greeks, which is why the Rannacher steps are kept by default. In space the centred scheme is second order. With $n_x = 501$, $n_t = 1000$ and Crank–Nicolson–Rannacher the price is accurate to a few 10^{-7} (space error) and costs a few tens of milliseconds in the browser.

This study also caught an error in an early version of the port: the θ-scheme used $\eta(t_n)$ in both halves of the step, a first-order error in time since $\eta(t) = \sigma(t)e^{\lambda t}$ varies within a step. The scheme now uses $\eta(t_n)$ in the implicit part and $\eta(t_{n+1})$ in the explicit part, with σ read on the segment of the step.

6.4 Exercise boundary

At each date the holder exercises where $h_k(x) > C_k(x)$. For a payer this is a half-line $x > x_k^\star$: high factor, high rates, valuable payer swap. Figure 2 shows C_k and h_k at T_1 and T_2; at the last date $C_m = 0$ and the option is exercised as soon as it is in the money.

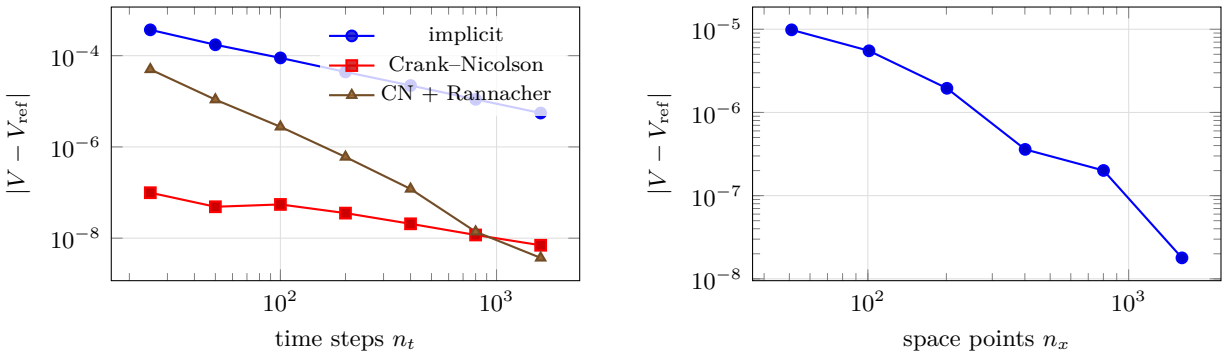


Figure 1: PDE convergence on the reference case. Left: time error ($n_x = 1001$). Right: space error (CN + Rannacher, $n_t = 4000$, reference $n_x = 4001$).

n (standard deviations)	Bermudan price
2	0.01675392
2.5	0.01674738
3	0.01674689
3.5	0.01674687
4	0.01674687
5	0.01674687
6	0.01674687
7	0.01674687

Table 1: Truncation of the PDE domain to $\pm n\sqrt{\alpha(T_m)}$ (space step kept constant).

7 Longstaff–Schwartz Monte Carlo

7.1 Algorithm

The factor is simulated exactly: $X_{t+\Delta} - X_t \sim \mathcal{N}(0, \alpha(t+\Delta) - \alpha(t))$, independent of the past. The continuation value in (13) is a conditional expectation, estimated by least squares (Longstaff and Schwartz 2001).

Algorithm 1 Longstaff–Schwartz for the Bermudan swaption

simulate N paths of X (and of $I_t = \int_0^t r_s \,\mathrm{d}s$) at the exercise dates
$\mathrm{CF}^{(i)} \leftarrow h_m(X^{(i)}_{T_m})$, $\tau^{(i)} \leftarrow T_m$ for every path i
for $k = m-1, \dots, 1$ **do**
  $\mathcal{I} \leftarrow \{i : h_k(X^{(i)}_{T_k}) > 0\}$ ▷ in-the-money paths only
  regress $Y^{(i)} = e^{-(I^{(i)}_{\tau^{(i)}} - I^{(i)}_{T_k})}\,\mathrm{CF}^{(i)}$ on $\{1, \tilde{x}, \dots, \tilde{x}^d\}$, $\tilde{x} = x/\sqrt{\alpha(T_k)}$, over $i \in \mathcal{I}$, giving $\hat{C}_k$
  for $i \in \mathcal{I}$ with $h_k(X^{(i)}_{T_k}) > \hat{C}_k(X^{(i)}_{T_k})$ **do**
    $\mathrm{CF}^{(i)} \leftarrow h_k(X^{(i)}_{T_k})$, $\tau^{(i)} \leftarrow T_k$
  end for
end for
return $\hat{V}_0 = \frac{1}{N}\sum_i e^{-I^{(i)}_{\tau^{(i)}}}\,\mathrm{CF}^{(i)}$ and its standard error

Normalising the regressor by $\sqrt{\alpha(T_k)}$ does not change the fit but keeps the normal equations well conditioned; the least-squares problem is solved by a Householder QR factorisation.

7.2 The two biases

Write $\hat{\tau}$ for the exercise rule produced by the regression.

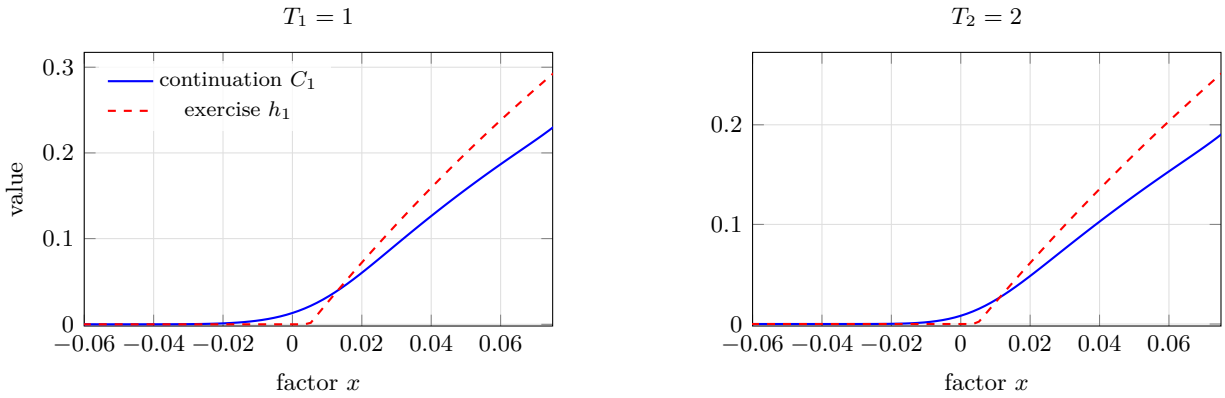


Figure 2: Continuation and exercise values from the PDE. The exercise boundary is the crossing point.

- *Low bias.* For any stopping rule, $\mathbb{E}[\beta_{\hat\tau}^{-1} h_{\hat\tau}] \le V_0$. If $\hat\tau$ is estimated on one set of paths and applied to an independent set, the second set gives an unbiased estimate of this quantity, hence a *lower bound* of V_0 in expectation.
- *High bias.* When the same paths are used to estimate $\hat\tau$ and to price, the rule is fitted to the noise of those paths and anticipates their future, which pushes the estimate up.

The page offers both estimators. An upper bound can be obtained by duality (Rogers 2002; Haugh and Kogan 2004; Andersen and Broadie 2004), a natural extension.

7.3 Discounting

Discounting must use the path of the short rate: $\beta_{T_k}^{-1} = e^{-I_{T_k}}$. The implementation simulates X on a fine grid (52 steps per year by default) and integrates $r(t, X_t)$ by the trapezoidal rule, an $O(\Delta t^2)$ error. An exact alternative exists: (X_t, I_t) is a Gaussian vector whose mean and covariance follow from (7), so it could be sampled jointly at the exercise dates without any grid.

7.4 Results

Figure 3 shows the three estimators against the PDE price. With exact discounting, both the in-sample and the independent estimators contain the PDE value in their confidence intervals: with a degree-2 regression on one factor, both biases are smaller than the statistical error at these sample sizes. The original discounting (section 8) stays about 3 bp of notional above the PDE whatever the number of paths: it is a bias, not noise.

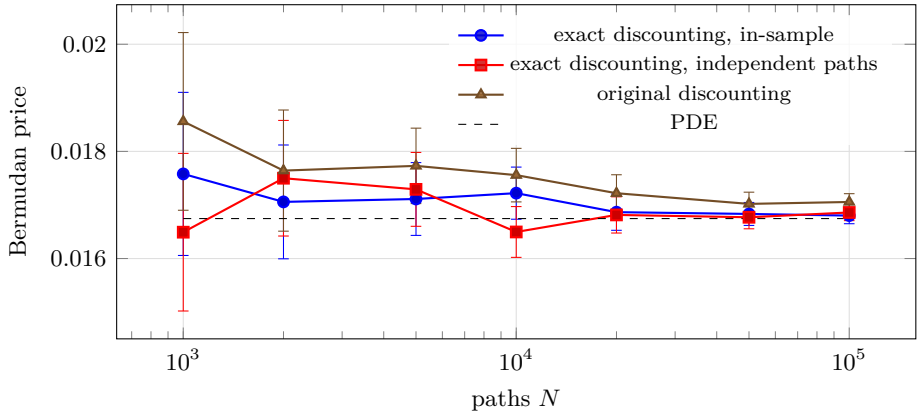


Figure 3: Longstaff–Schwartz estimates with 95 % confidence intervals, degree-2 regression, against the PDE price.

8 Review of the original model

8.1 A sign error in the convexity term

The original `hjm_1f.py` implements

$$P^{\text{code}}(t,T) = \frac{P(0,T)}{P(0,t)} \exp\Big(D(t,T)\big[c(t,T) - X_t\big]\Big), \tag{16}$$

that is (8) with $+c$ instead of $-c$, and its short rate, $r_t^{\text{code}} = -\partial_T \ln P^{\text{code}}(t,T)|_{T=t}$, inherits the error: $r_t^{\text{code}} = f(0,t) - e^{-\lambda t}\big[\bar{B}(t)\alpha(t) - \gamma(t)\big] + e^{-\lambda t}X_t$. Proposition 2 derives the correct sign directly from the drift condition. Because $c(t,T) > 0$ grows with $\alpha(t)$, the error grows with volatility and maturity.

Why calibration does not see it. Jamshidian's formula (proposition 4) uses $P(0,\cdot)$ and the variance v^2 of bond ratios, neither of which depends on c: Europeans are priced correctly and the calibrated σ are the same to 10^{-2} bp. The error only appears in methods that use $r(t,x)$ and $P(t,T;x)$ along paths, the PDE and the Monte Carlo, so the model prices the same European differently depending on the method.

8.2 The martingale test

Under $\mathbb{Q}$, $\mathbb{E}[\beta_t^{-1}P(t,T)]$ must equal $P(0,T)$ for every t. Figure 4 estimates it for $T = 7$ with 200 000 paths ($\sigma = 150$ bp, $\lambda = 3\,\%$). The original formula drifts away, to 2.2 % at $t = 6$; the corrected one stays within the confidence band.

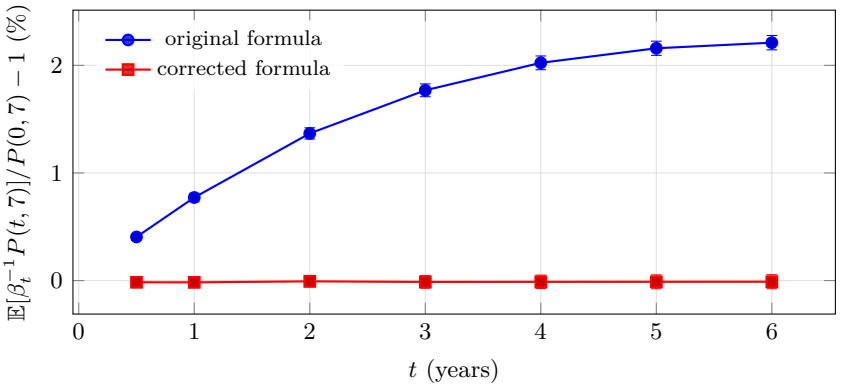


Figure 4: Martingale test of discounted bond prices, with 95 % confidence intervals.

8.3 Impact on prices

The last row is the most telling. Since Europeans are unaffected, the whole error falls on the early-exercise premium, the quantity a Bermudan model exists to price: the original formula underestimates it by 75 %.

8.4 Other findings

- *Monte Carlo discounting.* `mc_1f.py` freezes the short rate between exercise dates and discounts the exercise value back to today at $r(0)$, ignoring the curve: on the reference case, Monte Carlo bond prices are off by 0.26 % at 2 years and 0.59 % at 3 years, and the Bermudan is overpriced by about 3 bp of notional. Fixed by the exact discounting of section 7.

	original formula	corrected formula
$\mathbb{E}[\beta_3^{-1}P(3,7)]$, vs $P(0,7)=0.842000$	0.856942 (+1.78 %, 83 s.e.)	0.841949 (−0.3 s.e.)
European 3y → 7y, Jamshidian	0.02582	0.02582
European 3y → 7y, PDE	0.02057	0.02582
European 3y → 7y, Monte Carlo	0.02049 ± 0.00015	0.02574 ± 0.00017
Reference Bermudan (PDE, implicit)	0.014929	0.016737
Early-exercise premium	0.000601	0.002409

Table 2: Consequences of the sign error. Rows 1–4: $\sigma = 150\,\text{bp}$, $\lambda = 3\,\%$. Rows 5–6: reference case ($n_x = 501$, $n_t = 1000$); the premium is the Bermudan minus the most expensive European, 0.014328, which is the same with both formulas.

- *Regression bias.* Pricing on the regression paths only; independent paths and a choice of degree were added.
- *Time scheme.* Implicit Euler only; Crank–Nicolson with Rannacher steps was added (fig. 1).
- *Receiver swaptions.* The Python returns the payer formula for a receiver; the port uses bond calls.

9 Mean reversion under recalibration

9.1 Correlation between exercise dates

Since X has independent increments, $\operatorname{Cov}(X_s, X_t) = \alpha(s)$ for $s \le t$ and

$$\rho(s,t) = \operatorname{corr}(X_s, X_t) = \sqrt{\alpha(s)/\alpha(t)}. \tag{17}$$

The short rate and, to first order, the swap rates are affine in X with deterministic coefficients, so (17) is also their correlation across dates.

Proposition 6. *For a constant σ and $0 < s < t$, $\rho(s,t)^2 = \dfrac{e^{2\lambda s}-1}{e^{2\lambda t}-1}$ is strictly decreasing in $\lambda > 0$, from s/t at $\lambda = 0$ to 0 as $\lambda \to \infty$.*

Proof. $\partial_\lambda \ln \rho^2 = g(s) - g(t)$ with $g(u) = \dfrac{2u\,e^{2\lambda u}}{e^{2\lambda u}-1} = \dfrac{1}{\lambda}\,\psi(2\lambda u)$ and $\psi(z) = z/(1-e^{-z})$, which is strictly increasing on $(0,\infty)$. Hence $g(s) < g(t)$. The limits follow from $e^{2\lambda u} - 1 \sim 2\lambda u$ and from $\rho^2 \le e^{-2\lambda(t-s)} \cdot \frac{1}{1-e^{-2\lambda t}}$. □

9.2 Why the Bermudan becomes more valuable

Recalibration pins each European: in the swap-rate approximation, the k-th European fixes the model variance of its own swap rate, $\alpha(T_k)\,\kappa_k(\lambda)^2 \approx \sigma_{N,k}^2 T_k$, where $\kappa_k = \partial S_k/\partial x$ is a duration-like factor containing $e^{-\lambda T_k}$. A higher λ damps the impact of past shocks, so the calibration raises σ (fig. 5, left: from (77.9, 87.3, 96.7) bp at $\lambda = 0$ to (188.9, 214.4, 228.9) bp at $\lambda = 0.3$). The marginal law of each exercise value is then essentially unchanged, but the dates decorrelate: $\rho(T_1, T_3)$ falls from 0.513 to 0.349. With the marginals fixed and the dependence weaker, the right to choose the best date is worth more, exactly as the value of an option on the maximum of two assets increases when their correlation decreases. The Bermudan rises from 0.01656 to 0.01848 while every European stays at its market price (fig. 5, right).

Without recalibration (σ frozen at the $\lambda = 3\,\%$ values) the effect reverses: a higher λ only reduces the variance of every future rate, and every price falls.

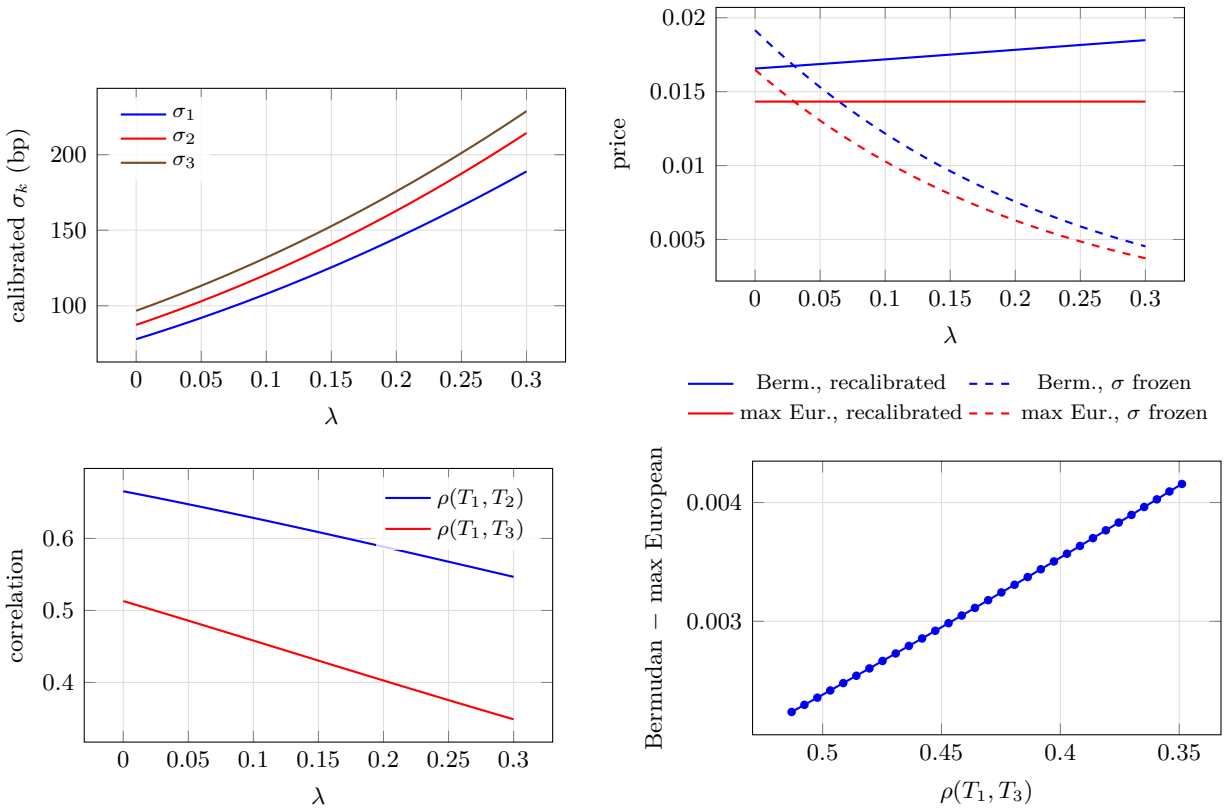


Figure 5: Mean reversion with recalibration. Top: calibrated volatilities and prices. Bottom: correlation between exercise dates and the early-exercise premium against it (λ from 0 to 0.3).

Remark 3. This is a tendency, not a theorem: it relies on the calibration set (co-terminal Europeans), on the payoff, and on the one-factor structure. In a two-factor model, correlation between dates can be lowered without changing λ, which is one reason why two-factor models are preferred for products sensitive to inter-date correlation (section 10).

10 A second factor

In one factor, λ is not identified by the co-terminal Europeans and it ties together the volatilities of the swaptions of all tenors, which are all driven by the same Gaussian variable. A second factor removes both limits. The price is that Jamshidian's decomposition no longer exists.

10.1 The model

Two Brownian motions W^1, W^2 with $d\langle W^1, W^2\rangle_t = \rho\, dt$ drive the forward rates with separable volatilities $\sigma_k(t,T) = \sigma_k(t)e^{-\lambda_k(T-t)}$, $k = 1, 2$; this is the two-factor Hull–White model, $r = r_0 + r_1 + r_2$ with $dr_k = -\lambda_k r_k\, dt + \sigma_k\, dW^k$, in which ρ is typically close to -0.9 (a steepening of the curve is negatively correlated with a parallel move) [10]. With $\rho_{kk} = 1$ and $\rho_{12} = \rho$, the model is Markov in the driftless Gaussian vector $X^k_t = \int_0^t \sigma_k(u)e^{\lambda_k u}\, \mathrm{d}W^k_u$, whose covariance is

$$A_{kl}(t) = \rho_{kl} \int_0^t \sigma_k(u)\sigma_l(u)\, e^{(\lambda_k+\lambda_l)u}\, \mathrm{d}u.$$

Let $D_k(t,T) = \bar{B}_k(T) - \bar{B}_k(t)$ and $G_{kl}(t) = \rho_{kl} \int_0^t \sigma_k \sigma_l e^{(\lambda_k+\lambda_l)u} \bar{B}_l(u)\,\mathrm{d}u$. The no-arbitrage drift of the forward rates gives

$$P(t,T) = \frac{P(0,T)}{P(0,t)} \exp\Big(- \sum_k D_k(t,T) X_t^k + \kappa(t,T) \Big),$$
$$\kappa(t,T) = -\tfrac{1}{2} \sum_{k,l} \Big[\big(\bar{B}_k(T)\bar{B}_l(T) - \bar{B}_k(t)\bar{B}_l(t)\big) A_{kl}(t) - D_k(t,T) G_{kl}(t) - D_l(t,T) G_{lk}(t) \Big]. \quad (18)$$

With $\sigma_2 = 0$, $\kappa = -D\,c$ and (18) is the corrected one-factor formula. Under the T-forward measure the factors keep a covariance $A(t)$ and acquire the deterministic mean $m_k(t;T) = -\sum_l \rho_{kl} \int_0^t \sigma_k \sigma_l e^{\lambda_k u} \bar{B}_l(T-u)\,\mathrm{d}u$. The martingale defect $\mathbb{E}^T[P(t,b)/P(t,T)] - P(0,b)/P(0,T)$ computed with these two ingredients is zero to machine precision for generic parameters.

10.2 What replaces Jamshidian

Jamshidian's trick needs the swap value to be monotone in the single factor, so that one root $x^\star$ splits the swaption into options on zero-coupon bonds. With two factors the swap value $S(x_1, x_2)$ is a function of the plane and the exercise set $\{S > 0\}$ is bounded by a curve: there is no common strike and no decomposition. The European price is the integral

$$V_0 = P(0,T_0)\,\mathbb{E}^{T_0}[S(X_{T_0})^+], \qquad X_{T_0} \sim \mathcal{N}(m(T_0;T_0), A(T_0)),$$

evaluated on a fine uniform grid of the normal law: since the payoff has a kink, such a rule is more accurate than Gauss–Hermite (relative error $4 \cdot 10^{-4}$ against Jamshidian in the one-factor limit, with 161×161 points). The bootstrap still works, because the price increases with the volatility level, but each Brent step now costs an integral.

10.3 Calibration: what is identified

For given $(\lambda_1, \lambda_2, \rho, \sigma_2/\sigma_1)$ the co-terminal bootstrap fixes the level of σ_1 exactly, but these structural parameters are invisible to the co-terminals. They are identified by swaptions of other tenors, through a global fit (Nelder–Mead on the root-mean-square gap in normal volatility, the level being re-bootstrapped at every evaluation).

We use the whole market of the ENSAE course [8] (EUR, 30 March 2016): the stripped curve, the 21×14 matrix of ATM normal volatilities (240 swaptions ending within 30 years), 16 ATM caps (1 to 30 years, Euribor 6M caplets, i.e. puts on zero-coupon bonds, closed form in one or two factors) and 13 swaption smiles. The Bermudan is a payer swaption into a 10-year swap, exercisable at 1 to 5 years, struck at the 10-year par rate (0.532 %). The model has **11 volatility segments**, one at every expiry of the matrix up to 5 years (1M, 2M, 3M, 6M, 9M, 1Y, 18M, 2Y, 3Y, 4Y, 5Y), each fitted exactly: on the co-terminal swaption *quoted along the smile at the Bermudan's strike* (it lies 8 to 53 bp below the forward swap rates; the smile interpolates the 13 tabulated smiles by a kernel in log-expiry and log-tenor and moves the quotes by up to 1.3 bp) or, for the other expiries, on an ATM swaption. The structural parameters are fitted on the **143 ATM swaptions and 6 ATM caps** that expire within the exercise window, and both models are then judged on all 240 swaptions and 16 caps. The bounds $\lambda_1 \in [-0.5, 0.5]$, $\lambda_2 \in [-0.5, 3]$, $|\rho| \le 0.95$ and $\sigma_2/\sigma_1 \in [0.05, 8]$ are imposed: with $|\rho|$ closer to 1 the factors become collinear and the exact bootstrap becomes unstable.

What the data say. (i) Mean reversion is negative: forward volatility *increases* with maturity (1-year cap at 21 bp, 1y×9y swaption at 60 bp). Fitted on swaptions alone a single factor finds $\lambda = -0.04$; fitted on the six caps alone it wants $\lambda \approx -0.15$ (RMSE 6.9 bp against 21.7 bp at -0.039): *one factor cannot agree with both markets*, which is the clearest case for the second factor. (ii) The second factor divides the error by 3.2 on the fitted swaptions and by 3.6 on the

	parameters	fitted swpt	fitted caps	beyond swpt	beyond caps	Bermudan
1 factor	$\lambda = -0.039$	12.4	21.7	21.4	7.6	0.05006
2 factors	$\lambda = (-0.003, 0.534)$, $\rho = -0.95, 1.48$	3.9	6.0	11.8	9.6	0.04816

Table 3: Root-mean-square error in bp of normal volatility on the fitted swaptions (143) and caps (6), and on those beyond the 5-year window (97 swaptions, 10 caps), price of the 10-year Bermudan and exercise premium (Bermudan − most expensive co-terminal European). The correlation sits on its bound.

fitted caps, and almost halves it beyond the window (21.4 to 11.8 bp); it is not better everywhere (caps beyond the window: 9.6 against 7.6 bp). (iii) The Bermudan is 3.8 % lower. (iv) With the window extended to 9 years the fit degrades (6.4 bp against 11.9) and the last segments become erratic: the model is reliable up to about 5 to 7 years.

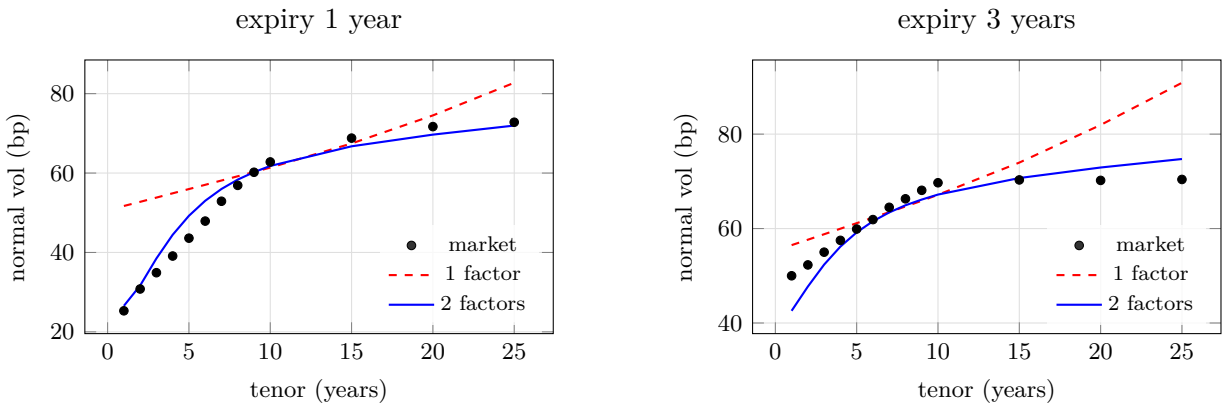


Figure 6: ATM normal volatility against the tenor of the swap, EUR, 30 March 2016 (expiries 1 and 3 years; all tenors up to 10 years are in the fit). With one factor the volatility is nearly flat in the tenor; the market, and the two-factor model, are steeply increasing at short expiries.

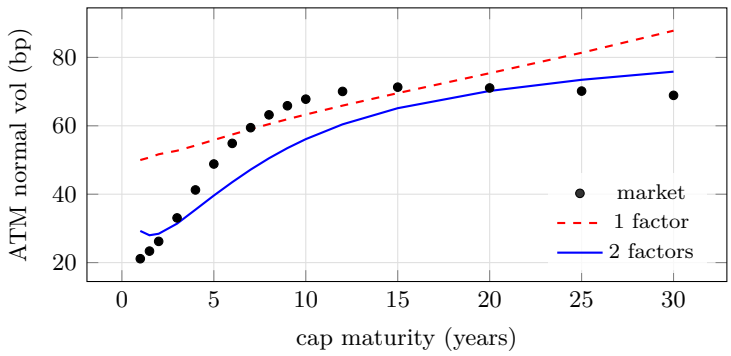


Figure 7: ATM cap volatilities (the first six maturities are fitted, the others are predicted).

Caveats. The correlation sits on its bound −0.95: the market asks for nearly collinear factors, a stretch of the model rather than a measured correlation. The residuals keep a structure (+6 bp at mid tenors for short expiries, −3 bp at long tenors): an exponential volatility profile has too little shape for this market. The objective has a flat valley and the individual parameters are not sharply identified. The smile is ignored by the models: it is only used to quote the co-terminals at the Bermudan's strike (the market normal vol at ±25 bp is 10 to 15 % above the ATM one at a 1-year expiry). Cap conventions are simplified (single curve, 0.5-year accruals). Not used: the cap

smiles, the multi-curve data of another date and the SABR calibrations. A third factor, a shifted diffusion or a stochastic volatility would be the next steps.

10.4 The Bermudan on a two-dimensional grid

In two factors the PDE has a cross-derivative, but between exercise dates the factors are exactly Gaussian, so time need not be discretised. Under the T_N-forward measure, on values deflated by $P(t,T_N)$ (martingales, hence no discount factor to simulate),

$$U_j(x) = \max\Big(\frac{h_j(x)}{P(T_j,T_N;x)},\ \mathbb{E}^{T_N}\left[U_{j+1}(X_{T_{j+1}}) \,\middle|\, X_{T_j}=x\right]\Big), \qquad V_0 = P(0,T_N)\,\mathbb{E}^{T_N}[U_1(X_{T_1})],$$

where $X_{T_{j+1}} - X_{T_j} \sim \mathcal{N}\big(m(T_{j+1};T_N) - m(T_j;T_N),\ A(T_{j+1}) - A(T_j)\big)$ independently of the past. The conditional expectation is a Gaussian kernel integrated against a bilinear interpolation of U_{j+1}; the only errors are the grid and the kernel (1.2 % at 41×61, 0.2 % at 101×141). Longstaff–Schwartz runs on the same deflated values with a two-variable polynomial basis and brackets the grid price from below.

The Europeans do not determine the Bermudan. On the real case ($\lambda_2 = 0.534$, $\sigma_2/\sigma_1 = 1.48$), every point of fig. 8 is recalibrated to the same 11 volatility segments, hence to the same Europeans; the exercise premium moves from 46 bp at $\rho = -0.95$ (the fitted value) to 102 bp at $\rho = +0.9$, and the Bermudan from 0.0483 to 0.0539 (12 %). The one-factor model with $\lambda = \lambda_1$ gives 0.0510 (73 bp). This is model risk made visible, and also the freedom that lets the model fit the rest of the market: the global fit is what selects ρ.

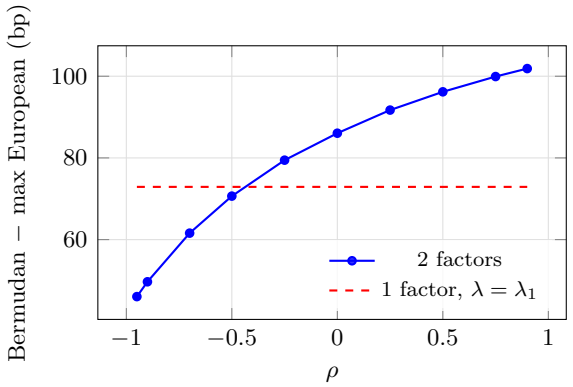


Figure 8: Exercise premium against ρ ($\lambda = (-0.003, 0.534)$, $\sigma_2/\sigma_1 = 1.48$), each point recalibrated to the 11 volatility segments of the real case.

10.5 What is lost, what is gained

10.6 Validation

(i) The martingale defect is zero. (ii) With $\sigma_2 = 0$ the European integral reproduces Jamshidian ($4\cdot10^{-4}$ relative) and the grid reproduces the corrected one-factor PDE (0.2 %). (iii) With $\rho = 1$, $\lambda_1 = \lambda_2$ and $\sigma_1 + \sigma_2 = \sigma$ the one-factor price is recovered. (iv) Grid and Monte Carlo agree within 0.4 % on four scenarios. (v) The closed-form bond put agrees with the two-dimensional integral of the same payoff, and the cap strike with the course workbook (0.5257 %, within 1 bp). (vi) The JavaScript engine of the page and the Python package `hjm2f` agree to 10^{-13} or better on prices, covariances, caps, smile, segments and bootstrap, and the global fit returns the same parameters and errors in both. (vii) Every number of this section is recomputed by `validate_2f.py`.

	1 factor	2 factors
European swaption	closed form (Jamshidian)	**lost**: 2-D integral, about 26,000 points
Bermudan engine	tridiagonal PDE, a few ms	**lost**: 2-D grid, 0.4–1.5 s; three factors would need Monte Carlo
Co-terminal calibration	exact bootstrap	still exact, but four structural parameters need a global fit (one bound active)
Fitted swaptions (143)	RMSE 12.4 bp	**gained**: 3.9 bp
Fitted caps (6)	21.7 bp; swaptions and caps ask for different λ	**gained**: 6.0 bp, same parameters
Correlation of forward rates	1 for all maturities	**gained**: free
Bermudan with the same Europeans	almost fixed by λ	**gained**: set by the dependence (model risk as well)
Hedging factors	parallel shift	**gained**: level and slope

Table 4: One factor against two.

11 Validation

The JavaScript engine of the page is a line-by-line port of the Python package. With the original formula and the notebook settings (implicit scheme, $n_x = 501$, $n_t = 1000$, ± 5 standard deviations) it reproduces every PDE price of the notebook to the eighth decimal, for all grids, domains, $\lambda \in [0, 0.3]$ and $K \in [2\,\%, 4\,\%]$, and the calibrated σ to 10^{-17}. With the corrected formula, Jamshidian, the PDE and the Monte Carlo agree on European swaptions, and simulated bond prices recover the curve within their confidence intervals.

12 Conclusion

The one-factor HJM model calibrates exactly and quickly to co-terminal Europeans and prices Bermudans accurately by PDE, with Monte Carlo as an independent check. The review shows the value of model-level tests: a convexity sign error invisible to the calibration was caught by a martingale test and by the inconsistency between methods, and it moved the Bermudan price by 12 %. The remaining model risk sits in λ, which the calibration does not determine and which moves the Bermudan by 12 % over a plausible range. The two-factor version trades the closed forms for integrals and a two-dimensional grid, and in return fits the whole market (240 swaptions and 16 caps) instead of its diagonal: on the real EUR data the error on the fitted swaptions falls from 12.4 to 3.9 bp and the Bermudan is 3.8 % lower. With the Europeans held fixed, the correlation alone moves the Bermudan by 12 %, and the fit to the rest of the market is what selects it. Limits: the correlation sits on its bound, the model degrades beyond 5 to 7 years, and the smile is ignored. Natural extensions are a dual upper bound for the Monte Carlo, a third factor or a shifted diffusion for the negative-rate regime, and a stochastic volatility.

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